

Phase 1: Update User State

Exposure Saturation

$$D_t$$

Self-exciting Intensity

$$\lambda_{u,c}(t)$$

Category Seq

Items-categorized
Mapping

History $\mathcal{H}_t$

Select

Phase 2: Risk-Aware Category Selection

$$\Lambda(D_t)$$

$$\hat{R}_t^+(c)$$

Risk

Maximize Obj.
Eq.(12)

Reward



Rec



User

Phase 3: Feedback & Update

Update

$\mathcal{H}_{t+1}$

Pretrain